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Inequality in Permanent Income as a Determinant of Philippine Households' Savings

C. Francis C. Maliwat

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ABSTRACT

In this paper, I investigate how inequalities in permanent income affects the saving behaviors of Philippine households. I exploit the instrumental variable method to formalize an empirical model that enables me to determine how permanent and transitory income affects the household saving rate across subsamples of households. I find that, unlike upper-class households, lower- and middle-class households are likely to spend large enough portions of additional permanent income to reduce their saving rates. Meanwhile, all households regardless of income classification tend to save larger portions of additional transitory income. This suggests that households are driven to spend and to save contingent on how they perceive the permanency or transitoriness of additional income to be. I also find preliminary evidence that, on average, households do attempt to spend prudently as they spend more of their permanent incomes on essential and recurrent expenditure items such as those on food, housing, transportation, and communication. My results show the overarching contribution of permanent income in determining household saving and importance of policy interventions to improve the low-savings economic environment of the Philippines by at least revolving around raising permanent income levels or alleviating consumption-side pressures of households.

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Keywords : permanent income hypothesis, household saving, income inequality, instrumental variable method

Corresponding author : maliwatcc@bsp.gov.ph

* Central Bank Associate, BSP Research Academy

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1 Introduction

Philippine authorities and researchers have long been cognizant of low saving patterns among Philippine households. Under commission by the Bangko Sentral ng Pilipinas (BSP), Bersales & Mapa (2006) find that, from 1985 to 2003, the country recorded its lowest (absolute) household saving rate of 16 percent in 2003. Notably, they find that the trend of declining household saving rates persisted among households across all levels of income save for those in the highest income group. Terada-Hagiwara (2009) reports the same trend, noting that the country posted a national saving rate (as a share of gross domestic product) of 2 percent in 2007, a far cry from the 10 percent recorded in 2002.

Though marginal improvements have been seen since then, the household savings situation in the country remains to be an area of policy concern. The BSP identifies the low-savings environment to be one of the primary forces that hinders financial inclusion in the country. In its report on the results of the 2019 Financial Inclusion Survey, the BSP (2021) finds that almost half (47 percent) of Philippine adults did not have savings in the reference period. The lack of money (or income) was cited as the foremost reason for the inability to save.

Uncoincidentally, the incidence of saving is higher among those who are employed, those who have family members working abroad, and those who own their own businesses—those who have reliable sources of income. Saving incidence also increases along higher socioeconomic classes; around 79 percent of upper- to upper-middle-class households have savings, while only 48 percent (below half) of the poorest households have savings.

Clearly, the importance of income when it comes to household saving behaviors cannot be overstated. Investigating income-savings interplays among Philippine households thus is warranted.

In the literature of household income and saving, one oft-cited theory stands out in pertinence to this subject matter: the permanent income hypothesis (Friedman, 1957). The hypothesis posits that household income is made of two components: the permanent component, which is the expected average value of lifetime income streams; and the permanent component, which refers to unanticipated windfall income shocks. A key proposition of the hypothesis is that household consumption depends on permanent income alone. Thus, if households receive positive windfall incomes, they increase their savings. Conversely, if they experience downward income shocks, they dissave as they anticipate their incomes to return to their permanent path in the future.

In this study, I explore the consequences of the permanent income hypothesis on the saving behaviors of Philippine households under the context of stark income inequalities that plague the nation. I formulate an empirical model that uses the instrumental variable method to estimate permanent and transitory income components and to assess how these components determine the household saving rate across subsamples of households grouped according to income class.

Consistent with other studies in the literature, I find that investigating the income-savings relationship using observed measures of household income tends to overestimate how income affects saving, which potentially leads researchers to make erroneous conclusions about how households save when faced with additional income. The marginal effect of observed income on the household saving rate is dominated by the effect of the transitory component of income, rather than the permanent. If household savings were driven largely by transitory income, it is careless to conclude that households have the capacity to save regularly. This is particularly important when income inequality is considered as I find that lower- and middle-class households tend to spend large portions of additional permanent income to reduce their saving rates and less of transitory income shocks to increase the same measure.

I also find that additional permanent and transitory income both result in increased levels of spending. This is especially apparent for lower- and middle-class households. Notably, I find that the marginal propensities to consume out of permanent income among lower- and middle-class households are large enough to reduce their saving rates. In fact, the marginal propensity to consume out of permanent income is greater than unity for lower-class households. Meanwhile, the marginal propensities to consume out of transitory income are small enough such that households are able to increase their saving rates. Taken together, these results indicate that households are driven to spend and, conversely, to save based on precautionary motives; their perceptions about the permanency or transitoriness of income influence spending confidence.

I also find initial evidence that points to the existence of some degree of prudence in spending among households across all income classes. Lower- and middle-class households seem to use additional levels of permanent income for essential and regular expenditure items and transitory shocks for likewise necessary but less frequently incurred items.

My results show the overarching significance of permanent income when it comes to household saving. Data from official government surveys have all but confirmed this sentiment. Thus, policy interventions to improve the poor household savings situation in the Philippines should be geared towards increasing permanent income levels or relieving consumption pressures of households to effectively increase disposable permanent income. On the fiscal side, this entails preventing cost-push inflationary shocks, implementing developmental interventions to increase permanent income levels of households at the aggregate, and carefully setting consumption tax rates to prevent disproportionately affecting the poor. On the central bank's side, this means fulfilling the mandate of price stability and continuing financial education campaigns but with permanent income inequality considerations in mind.

The rest of the paper proceeds as follows. [Section 2](#) offers some supporting findings from extant literature on the relationship between household saving and income under the context of the permanent income hypothesis. [Section 3](#) describes the empirical strategy employed to quantitatively investigate the relationship of interest. [Section 4](#) discusses the data source and discloses a few particularities about how certain data

inputs should be defined and interpreted. [Section 5](#) reports and interprets the results of the empirical tests. [Section 6](#) discusses the policy and research implications of the findings. [Section 7](#) concludes.

2 Review of related literature

The centrality of income as a determinant of household savings is extensively investigated in contemporary literature on household savings. A popular strand of the literature deals with the question on how income inequality determines differences in household saving behaviors through the lens of the permanent income hypothesis (Friedman, 1957).

Authors investigating the income-saving relationship with the permanent income hypothesis in mind argue against using current (observed) income as a determinant of the household saving rate. They concur with Friedman (1957) in that there is the need to, in the words of Dynan et al. (2004), “distinguish those [households] with high lifetime income from those whose income is high only transitorily” and “to correct for measurement error in income”, because positive transitory income shocks and upward measurement errors will tend to overstate the effect of measured income on household savings.¹

Bebczuk et al. (2015) and Dynan et al. (2004), for example, demonstrate the presence of upward bias from transitory income and/or measurement errors by regressing household savings against current and permanent income individually in separate model specifications. They find that the effect of current income on household saving is generally greater than that of permanent income for countries in Latin America and the United States, respectively. This treatment of this particular matter is not new, of course, as even authors of much older studies such as Bautista & Lamberte (1990) and Rodriguez & Meyer (1988) perform tests of the same kind and find similar conclusions with regard to the presence of bias in marginal effects by using current measures of income instead of estimates of permanent income.

Isolating the effect of permanent income on household saving (versus that of current) is more significant when income inequality is considered. This is made evident in the studies of Zabet & Gomes (2021) and Gandelman (2017), whose results make it apparent that the upward bias of transitory income shocks and/or measurement errors is manifest mostly for higher-income households in Brazil and in the greater Latin American region, respectively. Their results also suggest the converse inference that the marginal effect of income on household saving is biased downward for lower-class households in the aforesaid regions when current income, rather than permanent, is used as the regressor.

¹ This is particularly important if I consider the income inequality aspect of household savings; I do not wish to erroneously conclude that households, especially those in the lower class, are capable of saving out of regular income when their reported savings are in fact driven mostly by chance income shocks at a given time.

Moreover, authors that traverse this route of analysis are interested in testing the core propositions put forth by Friedman (1957). In general, authors investigating this line of inquiry find themselves at odds with the proportionality hypothesis of the permanent income hypothesis, which proposes that the proportion of consumption with respect to permanent income is constant regardless of the level of income. For example, Dynan et al. (2004) in their influential work conclude decisively that “the rich do, indeed save more” (based on tests using both current and, more importantly, permanent income as determinants of household saving). More recent studies likewise find that households with greater wealth or higher expected lifetime incomes tend to save more than those without (Zabot & Gomes, 2021; Alvarez-Cuadrado & El-Attar Vilalta, 2018; Bozio et al., 2017; Gandelman, 2017). Reasons cited include bequest motives, precautionary saving, and even wealth incurrence merely for the sake of it (Carroll, 2000). Alan et al. (2015), however, dissent from the conclusions of the aforesaid group of authors as they find that richer Canadian households do not save more relative to those in the middle of the permanent income distribution. What they do find, though, is that the poor “in a lifetime sense” dissave. All things considered, the motivation to analyze the relationship between permanent income and household saving is established.

There are also authors of older studies who, intentionally or otherwise, challenge the strict assumption of the permanent income hypothesis that transitory income is wholly saved. Worth mentioning are the studies of Bautista & Lamberte (1990), Rodriguez & Meyer (1988), and Abdel-Ghany et al. (1983), who expressly investigate the effect of transitory income, in addition to permanent, on household saving and find that the marginal propensity to consume out of transitory income is not nil. The aforesaid authors also allude to income class considerations, albeit inexplicitly, as they study heterogeneities in the effects of permanent and transitory income components on the household saving rate based on differences in certain household characteristics that indicate in no small part the average income levels of households.

Rodriguez & Meyer (1988), for instance, find that the marginal propensity to save out of transitory income in fact is greater than that out of permanent income for rural Philippine households. Bautista & Lamberte (1990) report similar results and find evidence of heterogeneity in marginal propensities to save (both permanent and transitory income) contingent on differences in the regional location and on differences in urbanity of Philippine households. Meanwhile, Abdel-Ghany et al. (1983) likewise find that the marginal propensity to consume out of transitory income is greater than the marginal propensity to consume out of permanent income when transitory income is small relative to permanent income, opening a course for research that has to do with how the permanent-transitory makeup of income affects household saving.

That the Philippines is a country characterized by stark and persistent income inequalities and a low-savings macroeconomic environment makes this a crucially pertinent subject worth investigating. It is thus only fitting that I join other studies within this important realm of inquiry.

3 Empirical strategy

My study investigates two levels of inquiry regarding the saving behaviors of Philippine households. At the surface, I conduct a general analysis with the propositions of the permanent income hypothesis in mind. I then look deeper into the matter by incorporating income class considerations into the analysis.

In assessing how permanent income affects household consumption and saving behaviors, the most immediate complication has to do with how permanent income is estimated in the first place. The instrumental variable method is used extensively in this strand of the literature to address the problems associated with measurement errors and transitory shocks contaminating observed income. This approach enables me to estimate the permanent incomes of households by isolating the variation in observed income associated with household characteristics that determine the lifetime income potentials and subsequently evaluate the marginal effect of permanent income on the household saving rate using these estimates.

I note that the instrumental variable method only enables me to estimate the effect of permanent income on the household saving rate; by design, it ignores the effect of transitory income on the same measure. I thus exploit the method only for the intermediate steps in my analysis, and I rely on conceptual definitions proposed by the permanent income hypothesis to formalize a more complete empirical model that estimates how transitory income determines the household saving rate.

In formalizing the empirical model, I begin with the relevant response variable, which is this the household saving rate. This is mathematically expressed as

$$S_i = \frac{Y_i - C_i}{Y_i} \quad \text{Eq. 1}$$

where, for each household i , the household saving rate S is a function of observed income Y and observed consumption C . More exactly, the household saving rate is the ratio of excess income after expenditures with respect to total income.

The permanent income hypothesis proposes that income can be decomposed into permanent and transitory components. I can thus formally express this sentiment as

$$Y_i = \hat{Y}_i + e_i^Y \quad \text{Eq. 2}$$

where, for each household i , observed income Y is equal to the sum of permanent income $\hat{Y}$ and transitory income e^Y .

The hypothesis likewise proposes that consumption also has permanent and transitory components. This sentiment is similarly expressed as

$$C_i = \hat{C}_i + e_i^C \quad \text{Eq. 3}$$

where, for each household i , total household expenditures C is equal to the sum of permanent consumption $\hat{C}$ and transitory consumption e^C .

By virtue of substitution, Eq. 3 then becomes

$$S_i = \frac{(\hat{Y}_i + e_i^Y) - (\hat{C}_i + e_i^C)}{\hat{Y}_i + e_i^Y} \quad \text{Eq. 4}$$

which shows that the household saving rate is a function of permanent and transitory components of income and consumption.

All of the relevant component-determinants of the household saving rate are estimated starting with permanent income. I use the following income regression equation

$$Y_i = \alpha^Y + Z_i\zeta + X_i\xi + \epsilon_i^Y \quad \text{Eq. 5}$$

which states that, for every household i , income Y is a function of an instrumental variable Z , a vector of control covariates X , and an error term ϵ^Y . The Greek letters ζ and ξ are coefficient parameters for Z and X , respectively; while α^Y is the constant term.

From the income regression equation, I get an estimated equation in the form of

$$\hat{Y} = \hat{\alpha}^Y + Z_i\hat{\zeta} + X_i\hat{\xi} \quad \text{Eq. 6}$$

where the hatted Greek letters $\hat{\alpha}^Y$, $\hat{\zeta}$, and $\hat{\xi}$ are estimates of the true values of the non-hatted counterparts. Using these coefficient estimates, I derive fitted values for income $\hat{Y}$, which is my estimated measure of permanent income for each household i . These fitted values are used to compute three important estimates: transitory income, and permanent consumption and, as an automatic consequence, transitory consumption.

Because observed income is the sum total of permanent and transitory income, transitory income is simply the difference between my permanent income estimate and observed consumption. This is mathematically expressed as

$$e_i^Y = Y_i - \hat{Y}_i \quad \text{Eq. 7}$$

where, for every household i , the residual e^Y is my estimate for transitory income, which captures the variation of income not explained by my instrumental variable Z and other salient household characteristics contained in the vector X .

Following basic economic intuition, I assert that consumption is a function of income. I represent this assertion by the expression

$$C_i = \alpha^C + Y_i\nu + X_i\Xi + \epsilon_i^C \quad \text{Eq. 8}$$

which states that, for every household i , consumption is a function of income Y , the same set of controls in vector X , and the error term ϵ^C . The Greek letters ν and Ξ are coefficient parameters for Y and X , respectively; while α^C is the constant term for this regression equation.

However, the permanent income hypothesis states that permanent consumption is a function of permanent income alone; transitory income does not factor into permanent consumption decisions. To measure only the permanent portion of consumption, I substitute observed income Y with my estimates for permanent income $\hat{Y}$ (as is

normally done under the two-stage least squares instrumental variable method), resulting in the expression

$$C_i = \alpha^C + \hat{Y}_i v + X_i \bar{\varepsilon} + \epsilon_i^C \quad \text{Eq. 9}$$

From the consumption regression equation, I get the resulting estimated equation

$$\hat{C}_i = \hat{\alpha}^C + \hat{Y}_i \hat{v} + X_i \hat{\varepsilon} \quad \text{Eq. 10}$$

which allows me to take fitted values representing permanent consumption $\hat{C}$. The hatted Greek letters $\hat{\alpha}^C$, $\hat{v}$, and $\hat{\varepsilon}$ are estimates of the true values of their non-hatted counterparts.

I similarly estimate transitory consumption as I do with transitory income by calculating the residuals e^C for every household i ; I subtract observed values for total current consumption C by my estimates for permanent consumption $\hat{C}$, as shown by the expression

$$e_i^C = C_i - \hat{C}_i \quad \text{Eq. 11}$$

From [Eq. 1](#) through [Eq. 4](#), I know that the household saving rate is a function of permanent and transitory components of income and expenditures. I express this as an econometric equation in the form of

$$S_i = \alpha^S + \hat{Y}_i \psi + e_i^Y \tau + \hat{C}_i \chi + e_i^C \kappa + \epsilon_i^S \quad \text{Eq. 12}$$

where α^S is the constant term; ψ and τ are coefficients attached to $\hat{Y}$ and e^Y , respectively; χ and κ are coefficients attached to $\hat{C}$ and e^C , respectively; and ϵ^S is the error term of the household saving rate regression equation.² I interpret ψ and τ are the marginal effects of additional permanent and transitory income, respectively on the household saving rate, and χ and κ are the net marginal effects of additional permanent and transitory consumption, respectively, on the same measure.

Following Bautista & Lamberte (1990), Rodriguez & Meyer (1988), and Abdel-Ghany et al. (1983), I use residual values of income to represent the transitory incomes of households to determine how the measure affects the household saving rate. At this point, it is necessary to disclose that this approach seems to have fallen out of favor among more recent authors as most studies in the contemporary literature on the link between permanent income hypothesis and household saving do not test the effect of income residuals (representing transitory income) on household savings. Nevertheless, I justify the use of residual income to represent transitory income as a determinant of the household saving rate as I adhere to conceptual definitions proposed by the permanent income hypothesis.

Because I am using fitted values, rather than actual ones, to estimate [Eq. 12](#), the resulting standard errors will be inappropriate for interpreting the resulting significance of the variables; the standard errors will have to be adjusted. To do away

² Unlike other studies within this field, I do not control for household characteristics in the household saving rate regression equation. I am, in essence, using all of the covariates in X as additional instrumental variables to estimate both permanent income (in addition to Z) and permanent consumption.

with this complication, I resort to bootstrapping³ to yield approximate estimates for the asymptotically correct standard errors.

I know that extant literature has confirmed that, in general, higher-income households tend to save more of their permanent incomes. In mathematical terms, the relationship between permanent income and household saving nonlinear. I thus test whether this holds true in the Philippine context.

To account for class differences in Philippine households' saving behaviors, I test the empirical model on subsamples of households grouped according to income class⁴. In doing so, I control for heterogeneities in household characteristics that are determined by income class and variations in income bases that affect marginal effect estimates because of mathematical consequence. I also confirm the nonlinearity of the relationship between income (both permanent and transitory) and household saving by testing a modified version of the empirical model that includes square terms for permanent and transitory income to provide additional strength to my findings.

4 Data particulars

For my empirical analysis to be consistent with the conceptual definitions provided by the permanent income hypothesis, the data I use as inputs to my empirical model should conform with my analytical assumptions. I thus dedicate this section to discuss my data needs.

My data comes from the 2015⁵ iteration of the Family Income and Expenditure Survey (FIES) of the Philippine Statistics Authority (PSA). The FIES series consists of surveys geared towards measuring family income and expenditures for policymaking purposes of the national government (PSA, 2020).

My interest in this particular data set lies in the richness of data on income, its components, and sources; and on expenditures, their components, and categories; household characteristics; and demographic information; among others. The 2015 FIES also covers a sample of 41,544 households, providing me a sufficiently representative account of the Philippine population.

The FIES provides data on household income according to type and source. I do away with imposing arbitrary notions of permanency or transitoriness of income based on types and sources and consider the sum total of all incomes as my primary measure of observed income. In doing so, I allow my estimation strategy to estimate permanent income from this aggregate measure without prejudice to the aforesaid factors.

³ I perform 1,000 bootstrap replications for all empirical tests.

⁴ For my main analysis, I use subsamples grouped according to the income class definitions put forth by Albert et al. (2018). I provide their definitions in [Annex A, Table 9](#).

⁵ The 2018 iteration, though it is more recent, is not used on account of the seeming presence of measurement or encoding errors in income variables.

In measuring household consumption, I account for expenditures in cash only. The FIES data set provides for expenditures in kind and as gifts, but analysis finds no need for these kinds of expenditures. I consider cash expenses to be the only kind that is taken from household income to determine the level of household saving. For example, a household whose consumption is entirely in kind or gift-based will have a saving rate of 100 percent being that none of its income is ever used for expenses.

The income and expenditure data used are scaled downward by a factor of 10,000 for analytical brevity.⁶ As such, a unit increase in income or expenditures refers to an increase of ₱10,000 for either measure.

As many authors have done before me, I use the college educational attainment of the household head as my primary instrumental variable. The literature has accepted the household head's education to be strongly correlated with lifetime income. I assert that this holds true especially so in the Philippines. Olfindo (2018) and Acosta et al. (2017), for example, find that having a college degree is strongly correlated with higher wages and formal employment in the Philippines. Olfindo (2018), in particular, finds evidence that Philippine wage workers with a tertiary diploma are paid disproportionately higher than those without; additionally, the effect of possessing a tertiary diploma does not diminish with work experience.

Of course, I recognize that the robustness of my results hinges on the reliability of my permanent income estimates, which, in turn, are sensitive to the choice of instrument I use (Alan et al., 2015). I therefore conduct additional tests using various other instruments suggested in the literature to see if my general findings hold even as I alter the key inputs to the empirical model.

Dynan et al. (2004) offer three other instruments in addition to the education of the household head: household consumption, lagged labor income, and future labor income. Zobot & Gomes (2021) and Alan et al. (2015) also consider total consumption (Alan et al., 2015) and nondurable consumption (Zobot & Gomes, 2021; Alan et al., 2015). Because my data set does not contain historical nor future expectations data on income, I am limited to exploring categories of consumption as alternative instruments.

Bebczuk et al. (2015) use the ownership of real assets, particularly the ownership of homes and automotive vehicles as additional instruments. I follow their reasoning as intuition tells me that the ability to make big-ticket purchases are likely to be dependent on the level of permanent income flows. Households will generally rely on increases in expected lifetime income rather than wait for transitory income shocks to invest in such real assets that will be utilized for long periods of time. Of course, I do not discount that considerable windfall income shocks also enable households to purchase such real assets; however, if I posit that households generally tend to use chance income shocks to purchase such real assets, then I should find that only those fortunate enough to experience windfall income will possess houses and automotive vehicles. I note that, because of the presence of missing observations in the variable

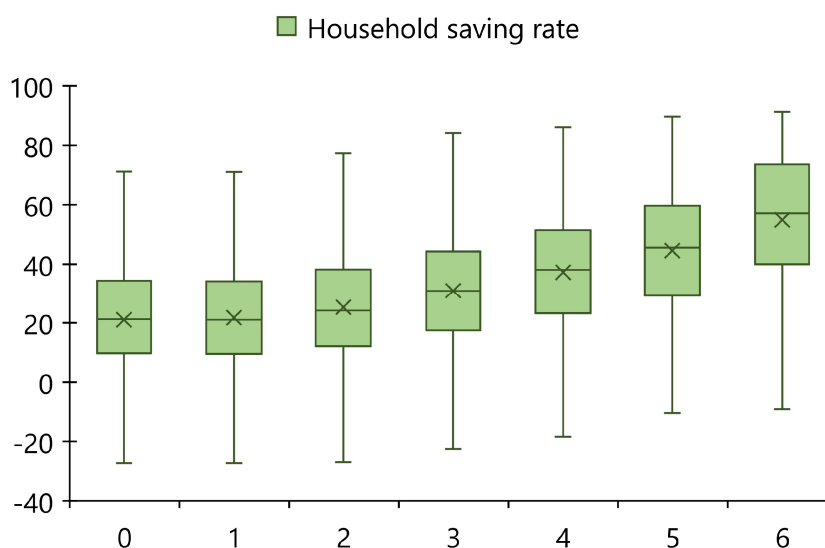
⁶ The scale factor is chosen based on the minimum value of income in the data set (₱11,285), which is in the ten thousands.

pertaining to the ownership of automotive vehicles, I am limited to using home ownership only as my additional instrument.

5 Results and discussion

To begin my analysis, I take a preliminary look at how household saving rates differ across income levels (Figure 1). I confirm that the household saving rate is, on average, increasing at higher levels of observed income.

Figure 1 Permanent-transitory income compositions across income classes



Units in percentage terms. Outside values not displayed.

Source: Author's estimations.

I consider regressing the household saving rate on observed measures of income and consumption, first, and then on estimates of permanent and transitory components of income and consumption. I contrast the results of each test against the other in Table 1.

Table 1 Marginal effects of observed, and permanent and transitory income on the household saving rate

		(1) Observed <i>N</i> = 41,544	(2) Estimated <i>N</i> = 41,544
Income	Permanent	0.557 ***	2.061 ***
	Transitory		0.522 ***
Consumption	Permanent	-0.899 ***	-2.999 ***
	Transitory		-0.823 ***

Bootstrapped standard errors not reported.

* $p < 0.10$,

** $p < 0.05$,

*** $p < 0.01$

Source: Author's estimations.

Two things are immediately apparent. First, the magnitude of the estimated marginal effects of observed, total current (i.e., permanent and transitory) income on the

household saving rate lies between that of each of the respective isolated components of both measures. Second, the resulting estimate of the observed version seems to be much closer to that of the transitory component.

Table 2 Observed, permanent, and transitory income correlations

	Y	$\hat{Y}$	e^Y
Y	1.000		
$\hat{Y}$	0.404	1.000	
e^Y	0.915	0.000	1.000

Source: Author's estimations.

I make the same inferences about the correlations displayed in Table 2 another way. I know that the magnitudes of the marginal effects of permanent income and consumption are greater than those of the marginal effects of the transitory counterparts. I can determine that this is resultant from how the compositional

makeup of income and consumption relate with how large the observed income and consumption levels are. I prove this mathematically beginning with

$$S_i = \frac{(\hat{Y}_i + e_i^Y) - (\hat{C}_i(\hat{Y}_i) + e_i^C)}{\hat{Y}_i + e_i^Y} \quad \text{Eq. 13}$$

which is a condensed expression of the model I formalized using Eq. 1 through Eq. 12 previously. Eq. 13 only means that, for every household i , the household saving rate S is a function of permanent income $\hat{Y}$; transitory income e^Y ; permanent consumption $\hat{C}$, which, in turn, is a function of permanent income $\hat{Y}$; and transitory consumption e^C .

I first consider taking the marginal effect of additional permanent income $\hat{Y}$ on the household saving rate S . I translate this into mathematical terms as taking the partial derivative of S with respect to $\hat{Y}$, resulting in

$$\frac{\partial S_i}{\partial \hat{Y}_i} = \frac{-(\hat{Y}_i + e_i^Y) \frac{\partial \hat{C}_i}{\partial \hat{Y}_i} + (\hat{C}_i(\hat{Y}_i) + e_i^C)}{(\hat{Y}_i + e_i^Y)^2} \quad \text{Eq. 14}$$

The first term in the numerator, which is the product of observed income and the marginal propensity to consume (out of additional permanent income) turned negative, dictates that how the household saving rate changes with permanent income is also contingent on how permanent consumption changes with permanent income. But because I am holding permanent and transitory consumption constant by including these measures as regressors in the baseline empirical model, I modify Eq. 14 to get

$$\left. \frac{\partial S_i}{\partial \hat{Y}_i} \right|_{\frac{\partial \hat{C}_i}{\partial \hat{Y}_i} = 0} = \frac{(\hat{C}_i(\hat{Y}_i) + e_i^C)}{(\hat{Y}_i + e_i^Y)^2} \quad \text{Eq. 15}$$

which is simply observed consumption over the square of observed income.

I now consider taking the marginal effect of additional transitory income e^Y on the household saving rate S . I take the partial derivative of S with respect to e^Y this time to get the expression

$$\left. \frac{\partial S_i}{\partial e_i^Y} \right|_{\frac{\partial \hat{C}_i}{\partial \hat{Y}_i} = 0} = \frac{(\hat{C}_i(\hat{Y}_i) + e_i^C)}{(\hat{Y}_i + e_i^C)^2} \quad \text{Eq. 16}$$

which is exactly equal to the right-hand side of the expression in [Eq. 15](#).

I suppose two households incur the same level of increase in their incomes. I suppose further that they have the same level of consumption greater than nil. If both households also have the same level of total income, then the change in their respective household saving rates will be equal. At this point, the compositional makeup of income for each household does not matter, and which component of income increases for which household is inconsequential. However, if their initial incomes are unequal to begin with, the change in their respective household saving rates from the same increase in income will likewise be unequal; the household with a smaller income base will enjoy a larger change in its household saving rate.

On account of this mathematical imposition, and in conjunction with the results displayed in [Table 1](#), I can therefore infer that households whose incomes are mostly permanent in composition have lower income bases relative to those whose incomes are mainly transitory in nature. The converse inference is that households whose incomes are mostly transitory have higher income bases than those whose incomes are largely permanent. These are the same inferences I derived from the correlations displayed in [Table 2](#).

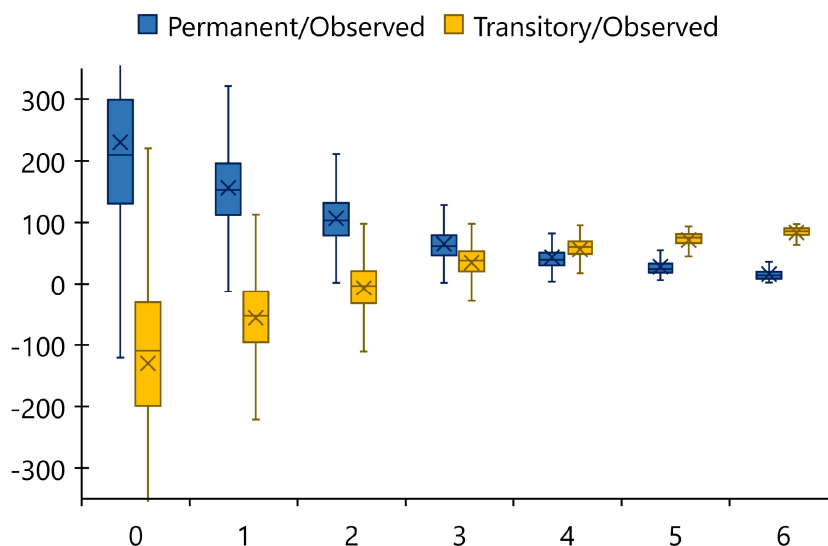
[Figure 2](#) shows obvious patterns on how the average income compositions⁷ of households change across income classes. On average, the proportion of estimated permanent income with respect to observed income decreases across higher income classes; conversely, the ratio of estimated transitory income to observed income increases. Notably, that the ratio of estimated transitory to observed income is, on average, mostly negative for lower-class households, suggesting that observed income measures for the subsample are, in fact, understated relative to what these households should expect their income levels to be given their characteristics.

The foregoing points establish the motivation to conduct this inquiry with income inequality in mind. I see that that difference in income bases render on-average analyses on what determines household saving rates quite meaningless from a more practical standpoint. As such, I am remiss to neglect accounting for income inequality in my analysis, more so given that the Philippines is a country characterized by stark income inequalities. I therefore conduct additional tests wherein income inequality

⁷ These income compositions are calculated using the estimates of permanent and transitory income per household based on the conceptual definitions proposed by the permanent income hypothesis. For a large number of households, estimated permanent income exceeds observed income, resulting in permanent to observed income ratios greater than unity. For the same households, transitory income is negative, resulting in negative transitory to observed income ratios. There are also a small number of households whose permanent incomes are negative and transitory incomes exceed observed income, resulting in permanent to observed income ratios less than nil and transitory to observed income ratios greater than unity.

takes center stage by estimating the empirical model on subsamples designated according to income class and note the following findings (Table 3).

Figure 2 Permanent-transitory income compositions across income classes



Units in percentage terms. Outside values not displayed.

Source: Author's estimations.

Table 3 Marginal effects of observed, and permanent and transitory income on the household saving rate across income class subsamples

		(1) Observed	(2) Estimated
LOWER CLASS		<i>N</i> = 26,341	<i>N</i> = 26,341
Income	Permanent	5.870 ***	5.829 ***
	Transitory		5.877 ***
Consumption	Permanent	-7.405 ***	-7.416 ***
	Transitory		-7.385 ***
MIDDLE CLASS		<i>N</i> = 14,725	<i>N</i> = 14,725
Income	Permanent	1.241 ***	1.561 ***
	Transitory		1.204 ***
Consumption	Permanent	-1.756 ***	-2.241 ***
	Transitory		1.687 ***
UPPER CLASS		<i>N</i> = 478	<i>N</i> = 478
Income	Permanent	0.104 ***	0.121 ***
	Transitory		0.102 ***
Consumption	Permanent	-0.352 ***	-0.407 ***
	Transitory		-0.343 ***

Bootstrapped standard errors not reported.

Source: Author's estimations.

* $p < 0.10$,

** $p < 0.05$,

*** $p < 0.01$

I also see that the marginal effects of observed income on the household saving rate are closer in magnitude to the marginal effects of transitory income on the same measure across all income classes. As before, I infer that the marginal effect of observed income on the household saving rate is dominated by the effect of transitory income. Thus, even within subgroups of income, much of the variation in observed income is explained by the transitory component, as increasing values of observed income correspond with greater proportions of transitory income relative to observed income. Correlations of observed income with permanent and transitory income, respectively, confirm this (Table 4).

Table 4 Observed, permanent, and transitory income correlations across income classes

LOWER CLASS			
	Y	$\hat{Y}$	e^Y
Y	1.000		
$\hat{Y}$	0.518	1.000	
e^Y	0.855	0.000	1.000
MIDDLE CLASS			
	Y	Y	Y
Y	1.000		
$\hat{Y}$	0.271	1.000	
e^Y	0.963	0.000	1.000
UPPER CLASS			
	Y	Y	Y
Y	1.000		
$\hat{Y}$	0.296	1.000	
e^Y	0.955	0.000	1.000

Source: Author's estimations.

I note that the marginal effect of permanent income is less than that of transitory income for lower-class households but greater for middle- and upper-class households. I know that permanent income dominates the income composition of households at lower income levels; the compositions of income at different income base levels dictate that the magnitudes of the estimated marginal effects displayed in Table 3 should be the way they are. I also know that, on average, the transitory to observed income ratio increases with income and exceeds the permanent to observed income at around the upper-middle income level, causing the average marginal effect of observed income on the household saving rate to be driven mostly by transitory income, rather than

permanent.

I now consider estimating the marginal effects of observed income, and permanent and transitory income on the household saving rate without controlling for consumption. I do this to deliberately introduce omitted bias into the estimated marginal effects of observed, and permanent and transitory income on the household saving rate. In doing so, I identify to what extent income interrelates with consumption to determine the household saving rate.

Table 5 shows that my general findings remain unchanged. The magnitudes of the coefficient estimates decrease across higher income classes on account of income base effects. The magnitude of the effect of observed income on the household saving rate is more similar to the effect of transitory income, rather than permanent, on the same. I also confirm that, across the board, the magnitudes of the estimated effects of permanent and transitory income on the household saving rate have been reduced. I can therefore infer that both components of income are correlated with consumption, either permanent or transitory, in some way. I know that the bias in the estimates for

permanent income are resultant from the omission of permanent consumption (by imposition of the empirical model) rather than of transitory consumption (by definition of transitory consumption). I can also infer that bias in the estimates for transitory income come from the omission of transitory consumption (though the model does not dictate that they should be correlated in any way) and not of permanent consumption (again, because of the design of the empirical model).

Table 5 Biased marginal effects of observed, and permanent and transitory income on the household saving rate across income class subsamples

		(1) Observed	(2) Estimated
ENTIRE SAMPLE		<i>N</i> = 41,544	<i>N</i> = 41,544
Income	Permanent	0.130 ***	−0.069 ***
	Transitory		0.169 ***
LOWER CLASS		<i>N</i> = 26,341	<i>N</i> = 26,341
Income	Permanent	−0.004	−2.215 ***
	Transitory		0.808 ***
MIDDLE CLASS		<i>N</i> = 14,725	<i>N</i> = 14,725
Income	Permanent	0.190 ***	−0.123 ***
	Transitory		0.214 ***
UPPER CLASS		<i>N</i> = 478	<i>N</i> = 478
Income	Permanent	0.047 ***	0.086 **
	Transitory		0.043 ***
Bootstrapped standard errors not reported. Source: Author's estimations.		* <i>p</i> < 0.10,	** <i>p</i> < 0.05, *** <i>p</i> < 0.01

Correlation analyses confirm my inferences to be true (Table 6). The findings are consistent across the entire sample and across the income class subsamples. I also find that the income-consumption correlations are decreasing along higher levels of income, consistent with the idea that higher-income households spend smaller proportions and, conversely, save more of their incomes relative to lower-income ones. I also confirm that transitory income is more greatly correlated with observed consumption than permanent income is, consistent with my earlier findings that transitory income takes up increasingly larger portions of observed income at higher income levels.

In synthesis, additional permanent and transitory income induce greater permanent and transitory consumption, respectively, but to such extents that the former is greater than the other. This is consistent across all households regardless of income levels, though the effect gets progressively weaker at increasing levels of income, which explains why the magnitude change in the coefficient estimates for income is greatest for the subsample of lower-class households and lowest for the subsample of upper-class households. My results also disagree with the findings of Abdel-Ghany et al. (1983) as I find that the marginal propensity to consume out of transitory income becomes greater than that of permanent income at the proportion of transitory income relative to observed income increases.

Table 6 Income and consumption correlations across income classes

ENTIRE SAMPLE			
	Y	$\hat{Y}$	e^Y
C	0.835	0.504	0.690
$\hat{C}$	0.396	0.981	0.000
e^C	0.736	0.000	0.805
LOWER CLASS			
	Y	$\hat{Y}$	e^Y
C	0.799	0.566	0.591
$\hat{C}$	0.504	0.973	0.000
e^C	0.622	0.000	0.727
MIDDLE CLASS			
	Y	$\hat{Y}$	e^Y
C	0.810	0.275	0.764
$\hat{C}$	0.218	0.806	0.000
e^C	0.783	0.000	0.813
UPPER CLASS			
	Y	$\hat{Y}$	e^Y
C	0.390	0.062	0.389
$\hat{C}$	0.053	0.180	0.000
e^C	0.396	0.000	0.414

Source: Author's estimations.

increases the household saving rate, suggesting that these households are not compelled to spend large portions of additional transitory income.

The results suggest that lower- and middle-class households feel more secure about spending additional permanent income than they do about doing the same for transitory income. Perceptions about the stability of additional income flows seem to matter for lower- and middle-class households. Terada-Hagiwara (2009) concludes similarly about the effect of certainty of permanent income flows. She finds that the extension of the coverage of social security (which effectively increased the expected value of lifetime disposable income of households) changed the precautionary saving motive of Philippine households and empowered them to spend with less worry.

Meanwhile, upper-class households are able to increase their saving rates, even though they do increase their consumption when their incomes, whether permanent or transitory, increase. This shows that, for upper-class households, the marginal propensities to consume from permanent income and, indeed, from transitory income are not large.

I also find from my income class subsample regressions that the marginal propensities to consume out of permanent and transitory income are not constant nor increasing at higher levels of income. Obviously, additional permanent income will not necessarily result in additional consumption that decreases the household saving rate when taken

What really stands out in my new set of results is that the estimated marginal effects of permanent income on the household saving rate for the subsamples of lower- and middle-class households have become negative. This suggests that the downward bias from excluding consumption controls is large enough to completely eclipse the inherently positive effect of permanent income on the household saving rate.

I can therefore infer that, for lower- and middle-class households, additional permanent income, on average, results in additional permanent consumption large enough to reduce their saving rates. From Eq. 14, I know this happens only when the absolute value of the first term in the numerator is greater than the second. This occurs when the marginal propensity to consume out of permanent income is sufficiently large. In contrast, for the same sets of households, the same level of additional transitory income

ad infinitum as implied by the whole-sample estimation results in Table 5. In other words, the relationship between households' incomes and their saving rates is nonlinear, and I am remiss to neglect to incorporate this sentiment into the empirical model. I thus modify the empirical model to account for this apparent nonlinearity by assuming the function takes the quadratic form.⁸

Table 7 Accounting for nonlinearity in the effect of permanent and transitory income on the household saving rate

	(1)
	<i>N</i> = 41,544
Permanent income	-0.681 ***
Square of permanent income	0.004 ***
Transitory income	0.261 ***
Square of transitory income	-0.000 ***
Constant	33.069 ***

Bootstrapped standard errors not reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's estimations.

The results in Table 7 confirm my intuition: a parabolic relationship exists between the household saving rate and permanent income. Because I now know that that the household saving rate function is quadratic with respect to permanent income, I can expect a shift in the direction of the effect of permanent income on the household saving rate at a certain level.

I know that this function, when graphically represented, is an upward parabola, being that the coefficient estimate of the square term of permanent income is positive. Thus, additional permanent income decreases the household saving rate at a decreasing pace, and, after a certain level of additional income, increases the household saving rate at an increasing pace. Using the coefficient estimates displayed in Table 7 and the formula to calculate the x -value of the vertex of a parabola, I can estimate the level of additional permanent income that yields a positive change in the household saving rate, which is

$$-(-0.6808896) \div (2 \times 0.0043669) = 77.9602922 \quad \text{Eq. 17}$$

or around ₱780,000 in annual terms. Meanwhile, the coefficient estimate for the square of transitory income is negative, indicating that the parabola faces downward. Therefore, the household saving rate increases with transitory income but decreases after a certain level of transitory income. However, the magnitude of the coefficient estimate of the square term is practically nil (more precisely, -0.0002059), so we can dismiss the relationship between transitory income and the household saving rate as practically upwardly linear.

At this point, I reiterate that, without the foregoing information, I may be led to mistakenly conclude that additional current income (without disaggregating the effects of each component) always results in additional saving, which is hardly the case when I wish to speak of saving sustainably from regular income, not chance income shocks. It is clear that transitory income shocks and/or measurement errors in income mask the

⁸ See Wooldridge (2010) for a discussion on how to appropriately estimate a nonlinear model with endogeneity to avoid what he calls the "forbidden regression".

true effect of additional income on the household saving rate, particularly for lower-class households.

My findings have unfortunate implications for lower- and middle-class households. Their incomes are low or middling, mostly permanently so, and are consistently inevitably exhausted. The next question, then, has to do with why they choose to consume large portions of additional income rather than to save.

Before I answer this question, I first disclaim that my empirical strategy does nothing to capture more nuanced aspects about the financial behaviors of households. My findings say nothing about financial literacy or financial prudence and responsibility. It could very well be that a household's permanent consumption is largely wasteful, though I cannot discern this from my data. These more nuanced aspects are also subject to arbitrary notions and judgments that are difficult to measure quantitatively and purely objectively.

I address these shortcomings is to investigate where households spend their permanent and transitory incomes. This time, I regress observed, and permanent and transitory income on expenditure levels of different consumption categories. I report the results of this exercise in [Table 8](#).

Of course, even within consumption categories, consumption can be essential or excessive; still, I can, to some extent, rely on economic intuition and judge which of these categories are, in general, necessary or inessential and infer how responsibly or imprudently these households spend their permanent and transitory incomes.

First, I confirm that lower- and middle-class households do, indeed, spend more of their permanent incomes, rather than transitory. Notably, my findings indicate that the marginal propensity to consume out of additional permanent income is over **100** percent for lower-class households, confirming my earlier finding that they do reduce their saving rates as their permanent incomes increase. Meanwhile, they only spend around **69** percent of additional transitory income, likewise confirming my earlier finding that they do save portions of additional transitory income. As for middle-class households, I find that the marginal propensities to consume from permanent income and from transitory income are around **75** percent and **59** percent, respectively. It appears that a marginal propensity to consume out of permanent income of **75** percent is enough to reduce the household saving rate for this subsample of households.

The results for upper-class households are less conclusive when it comes to the marginal effects of permanent income.⁹ The coefficient estimates across all regressions are largely insignificant, suggesting that the effect of additional permanent income is statistically unlike from nil. Some of the estimates are even negative, which go against economic intuition. Meanwhile, the estimated marginal effects of transitory income on

⁹ This might be resultant from the small sample size for this subgroup of households. I perform additional tests using income terciles as income class categories to increase the number of observations in the subsample of higher-income households. The results of this exercise are reported in [Annex A, Table 11](#) and [Table 12](#).

different consumption categories across all regressions are positive, though, in general, less statistically significant.

Table 8 Effects of additional permanent and transitory income on different consumption categories across income classes

	(1) Lower class	(2) Middle class	(3) Upper class
TOTAL OBSERVED	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	1.085 ***	0.752 ***	0.087
Transitory income	0.686 ***	0.586 ***	0.170 ***
FOOD	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.603 ***	0.215 ***	-0.016
Transitory income	0.272 ***	0.114 ***	0.011 **
HOUSING	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.206 ***	0.219 ***	0.037
Transitory income	0.146 ***	0.129 ***	0.047 ***
COMMUNICATION	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.019 ***	0.036 ***	0.000
Transitory income	0.015 ***	0.023 ***	0.004 ***
TRANSPORTATION	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.069 ***	0.067 ***	-0.016 *
Transitory income	0.048 ***	0.049 ***	0.016 ***
EDUCATION	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.026 ***	0.011 **	-0.011
Transitory income	0.027 ***	0.044 ***	0.010 **
CLOTHING	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.015 ***	0.011 ***	0.002
Transitory income	0.016 ***	0.017 ***	0.004 **
NONDURABLES	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.018 ***	0.023 ***	0.010
Transitory income	0.012 ***	0.025 ***	0.016 ***
DURABLES	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.007 ***	0.002	0.068 **
Transitory income	0.022 ***	0.022 ***	0.001
HEALTH	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.011 ***	0.029 ***	-0.001
Transitory income	0.031 ***	0.039 ***	0.012
LEISURE	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.007 ***	0.005 ***	-0.003
Transitory income	0.005 ***	0.008 ***	0.004 **
SPECIAL OCCASIONS	<i>N</i> = 26,341	<i>N</i> = 14,725	<i>N</i> = 478
Permanent income	0.009 ***	0.006 **	0.010
Transitory income	0.023 ***	0.022 ***	0.006 *

Bootstrapped standard errors not reported.

* $p < 0.10$,

** $p < 0.05$,

*** $p < 0.01$

Source: Author's estimations.

For lower- and middle-class households, much of additional permanent and transitory income go to food and housing. Lower-class households also tend to spend a sizable portion of additional income on communication expenses (e.g., internet subscriptions, or prepaid or postpaid mobile cellular plans). In addition, both sets of households spend a small yet perceptible portion of additional permanent and transitory income on transportation expenses. For all these consumption categories, the marginal propensity to consume from permanent income is greater than that from transitory income. It bears noting that these consumption categories are arguably necessary for survival, routinary in nature, and are thus largely unavoidable, explaining the gap in the marginal propensities to consume out of permanent and transitory income.

There are four consumption categories in which additional transitory income results in greater spending than does additional permanent income: education, clothing, durables, and health. Expenditures of these kinds are, to various extents, likewise necessary, but less frequently incurred and (with the exception of clothing) usually much higher in value than day-to-day expenses such as those on food, housing, communication, and transportation. It thus makes sense for households to use windfall incomes on such big-ticket items.

Finally, expenditures on nondurables, leisure, and special occasions do not seem to increase by much with additional permanent and transitory income, suggesting that these items do not fall within the realm of priorities of lower- and middle-class households. These consumption categories are arguably inessential (leisure and special occasions more so than nondurables) and are also less frequently incurred. In addition, leisure products and services may also be quite expensive, making them luxuries that lower- and middle-class households cannot readily afford.

The results suggest that lower- and middle-class households do try to allocate additional incomes responsibly. I see that large portions of additional income, either permanent or transitory (but more so for the former than the latter) tend to go toward items that most households cannot live decently without. Unfortunately, apparent attempts at prudence notwithstanding, low permanent income bases coupled with high costs of necessities make it difficult for these households to incur savings sustainably.

Before I conclude this discussion, I must tackle a few issues regarding my empirical strategy. While the results of my empirical tests are useful and generally consistent with theoretical expectations and economic realities, I accept that there are theoretical and empirical limitations that constrain me from performing such analyses that are fully captures all economic phenomena surrounding the matter.

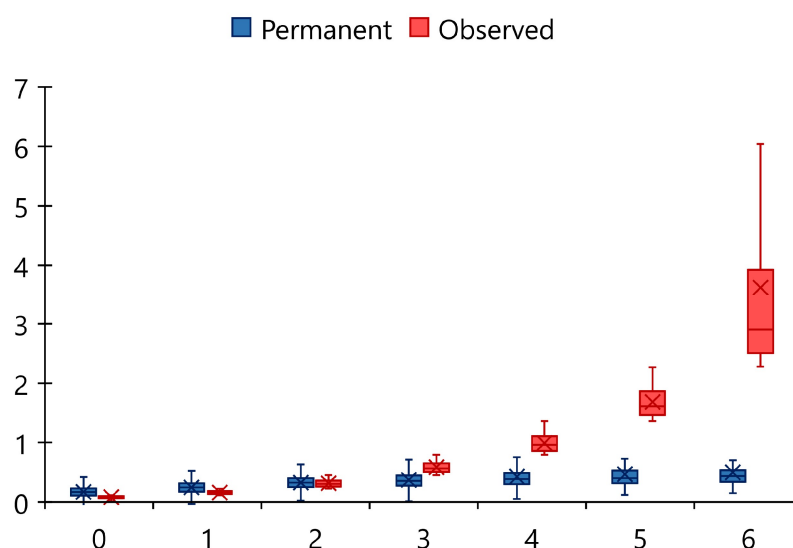
One such issue is that the predictive power of the empirical model seems to falter at increasing income levels. Gandelman (2017) expresses a similar sentiment in the conclusion of his study. I suspect that that the instrument I used, i.e., the college educational attainment of the household head, is not the strongest predictor of permanent income for wealthier households, hence the trends in income composition shown in [Figure 2](#). Intergenerational wealth persistence among upper class households should lead me to conclude that larger portions of their observed income

are more permanent than my estimates suggest. Privileges that come with wealth may also render the attainment of a college degree practically moot. Given these sentiments, I conduct additional tests using the alternate instruments discussed in [Section 4](#). I report the results of these supplemental tests in [Annex B](#).

Another issue has to do with the cross-sectional nature of the data and methodology used to perform the empirical tests. In this study, I capture variations across households to estimate permanent income, but I fail to account for one other factor central to the assumptions of the permanent income hypothesis: the time dimension. If historical income data per household were available, I could predict more expectations-based estimates of permanent income backed by past income trends. Panel data would also especially useful for estimating permanent incomes of households at higher income classes. With panel data, I could confirm whether upper-class households' incomes are consistently, not transitorily, high. Thus, panel data and methodologies that exploit such data are preferable under the context of this study.

[Figure 3](#) shows why I suspect that the empirical model's predictive power weakens along increasing income levels. The permanent income estimates begin to plateau somewhere along the middle-income level when estimating using the full sample. This indicates that there is a large portion in the variation in current income of upper-class households that our instrument fails to identify and capture.

Figure 3 Full-sample permanent income estimates vs. observed income

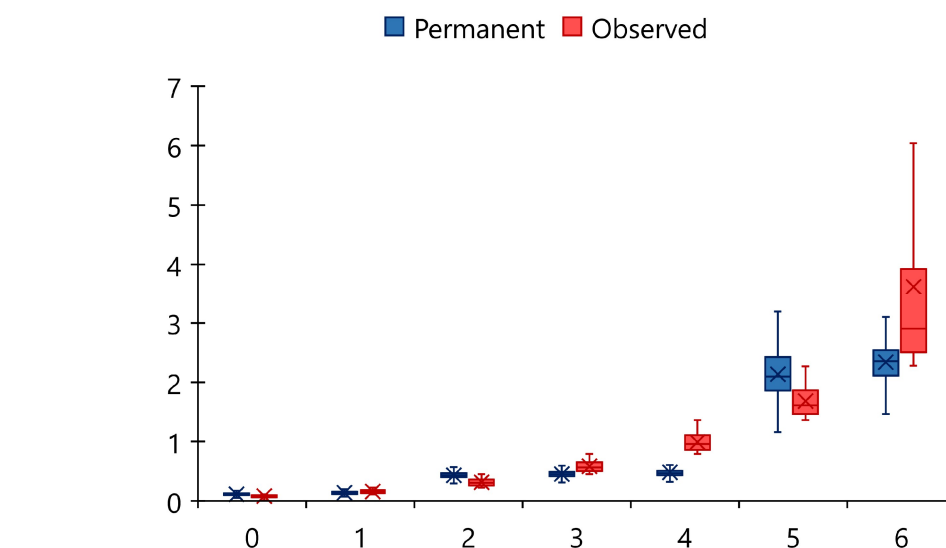


Units in million Philippine pesos. Outside values not displayed.

Source: Author's estimations.

Estimating permanent income by subsamples defined according to income class seems to rectify this problem to some extent as shown in [Figure 4](#).

Figure 4 Subsample permanent income estimates vs. observed income



Units in million Philippine pesos. Outside values not displayed.

Source: Author's estimations.

These issues notwithstanding, the value of my results is not diminished. My empirical strategy is founded on adherence to conceptual definitions, and my results do not defy basic economic realities. In any case, this study provides cause to reevaluate policies pertinent to the matter and a useful starting point for further research on the implications of the permanent income hypothesis on household savings in the Philippines.

6 Implications for policy and research

If income—permanent income, specifically—is so crucial to saving, policy interventions will have to revolve around raising permanent income levels or managing expenditure pressures of households to effectively increase permanent disposable income levels. It is also imperative for national government authorities to assess the policies explicitly geared towards promoting saving among households and those that only tangentially affect household saving by how they affect permanent income.

However, policymakers' immediate influence on household incomes is limited; unless macroeconomic circumstances make it feasible to provide universal basic income or to occasional and deliberately introduce transitory shocks to income to households, there is little they can do to increase the permanent income of households at an immediately tangible level. Developmental interventions to increase the permanent income of households, such as investing heavily in human capital development or creating more employment opportunities, may indeed allow households to break the permanency of poverty, but the returns on these investments take time to materialize.

Thus, Philippine policymakers must explore how to alleviate expenditure-side pressures of households to effectively raise the (permanent) disposable incomes of households at the aggregate level.

On the fiscal side, this means ensuring that product supply chains, especially for basic goods and services, are intact and undisrupted to prevent cost-push inflationary shocks. Fiscal authorities could also promote saving at a large scale by improving basic economic infrastructure and systems such as public education, socialized healthcare, and mass transportation, among others, to ease the financial burdens of households. They should also consider income inequality in raising consumption taxes, particularly those on basic commodities that are universally necessary (regardless of class); increasing the tax burdens on consumers will obviously be more damaging for poorer households whose incomes are already permanently low.

On the monetary side, promoting household saving entails maintaining low and stable inflation, particularly of basic commodities, by prudently managing aggregate demand through sound monetary policy. Though this is something the country's central monetary authority has already been doing by default, being that it is among the BSP's primary mandates, it bears noting as the impetus for carefully-calibrated monetary policy is emphasized.

The BSP also has more specific interventions to induce further household saving that are in line with its financial inclusion agenda. There are consumer-side interventions that aim to address shortcomings in demand for formal financial products and services. There are also supply-side interventions that rectify the lack of suitable financial products and services for those who remain to be unserved and underserved.

On the demand side, the BSP regularly conducts financial education and literacy campaigns to induce greater formal financial participation among Philippine citizens.¹⁰ In these campaigns, the BSP encourages participants to observe proper financial planning and budgeting; one such way is by promoting the "correct" saving formula, wherein a portion of income is allocated for savings first to determine expenditures (instead of savings being the residual of income and expenditures), in these campaigns. The BSP also educates them about available financial services offered by formal financial institutions.

The BSP in 2018 also issued circular introducing a framework that allows banks to offer basic deposit accounts to unserved and underserved consumers.¹¹ A basic deposit account features a "liberalized customer onboarding", low initial deposit and zero maintaining balance requirements, and no dormancy charges. In other words, a basic deposit account is designed to be free of transactional frictions at and beyond the account opening stage.

While all of the BSP's initiatives are undoubtedly welcome, there is cause to reevaluate the effectiveness of such interventions on the most vulnerable households given the overarching importance of permanent income to household saving. For example,

¹⁰ Copies of the BSP's financial education materials can be found in the central bank's official website.

¹¹ BSP Circular No. 992, series of 2018: Framework for Basic Deposit Accounts; dated 01 February 2018.

notions of a “correct” household saving formula may not be so appealing to households with low permanent income levels whose options are limited to saving or starving. There is also the recognition that a majority of Philippine households do not find the need for formal financial accounts primarily because they do not have the income to save in the first place; thus, even if low initial deposit and zero maintaining balance requirements may induce them to open basic deposit accounts, more vulnerable households may find limited opportunities to continually make deposits to increase their savings.

My results indicate that there are areas of opportunity for policymakers to promote household saving at the behavioral level, however. First, I find preliminary evidence of prudence among lower- and middle-class households with respect to how they spend their incomes. Second, I see that the aforesaid households are likely to increase their savings when they experience transitory income shocks, suggesting that their saving habits are driven by precautionary motives. It seems, then, that campaigns to enhance household budgeting and to encourage households to smooth their consumption may enable households to develop fruitful mindsets with regard to saving.

Along with existing ones in the literature, this study builds the case for the collection of more frequent and more exhaustive data to monitor the household saving situation in the Philippines. One such opportunity for enhanced data collection comes in the form of the Consumer Finance Survey (CFS) series of the BSP. The CFS series consists of more wealth-focused surveys that record more extensive accounts of household assets and liabilities in addition to current income and expenditure flows. These surveys have the potential to capture arguably better ideas of the permanent incomes of households. It would be considerably useful to design future iterations of the CFS with such theoretical frameworks as the permanent income hypothesis in mind. Additional variables that may serve as better instruments for permanent income will surely be welcome additions for future researchers to use and appreciate.

7 Conclusion

In this study, I assess Philippine household saving behaviors through the lens of the permanent income hypothesis (Friedman, 1957) and under the context of income inequality. The low-savings and high-inequality macroeconomic situation in the Philippines motivates my inquiry.

I formulate an empirical model based on the instrumental variable method to estimate permanent and transitory income and determine how these two income components affect the household saving rate across income classes. As do previous studies in the literature, I find that using observed income measures to test the income-saving relationship tends to overstate how income affects household saving. I also find that, based on my empirical tests, the observed incomes of lower- and middle-class households are mostly permanent, but those of lower-class households are, in fact, low relative to expected levels based on my estimates. Notably, I find that lower- and middle-class households tend to consume large portions of additional permanent

income for essential and regular expenses (suggesting prudence in spending) enough to actually decrease their saving rates.

My results indicate that inequalities in permanent income should be a central consideration when it comes to designing policies to encourage greater saving among Philippine households. There is also cause to reevaluate how effective existing policies are based on how they address or incorporate the implications of the permanent income hypothesis on household saving.

This study offers a preliminary analysis on the role of permanent income inequality on Philippine household savings, but much remains to be learned. In particular, there is a behavioral aspect to household spending and saving that this study cannot explore given certain data constraints. Because of the seemingly diminished predictive power of the model for upper-class households, there is also room for further analysis with regard to the saving behaviors of the rich. Thus, future research stands to benefit from enhancements in data collection. These limitations notwithstanding, this study provides value to government authorities and contributes to extant literature as it underscores the need to consider the poor in crafting policies and in conducting empirical inquiries.

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Annex A Supporting definitions and tests for income class analyses

In constructing the income class categories for the subsample regressions, I use the income class definitions specified by Albert et al. (2018) and the revised official poverty line P equal to ₱9,452 per month (or ₱114,504 per year) for a family of five in 2015 (PSA, 2019). For brevity in the analysis and interpretation of the results, the three higher-order classifications shown in Table 9 (lower class, middle class, and upper class) are used to designate households into their respective subsamples.

Table 9 Primary income class category definitions

Income class categorization	Definition	Indicative range
LOWER CLASS		
Poor	$Y < P$	[₱0, ₱114,504)
Low income but not poor	$P \leq Y < 2P$	[₱114,504, ₱229,008)
MIDDLE CLASS		
Lower-middle income	$2P \leq Y < 4P$	[₱229,008, ₱458,016)
Middle-middle income	$4P \leq Y < 7P$	[₱458,016, ₱801,528)
Upper-middle income	$7P \leq Y < 12P$	[₱801,528, ₱1,374,048)
UPPER CLASS		
Upper income but not rich	$12P \leq Y < 20P$	[₱1,374,048, ₱2,290,080)
Rich	$Y \geq 20P$	[₱2,290,080, ₱∞)

Indicative range figures are annualized.

Y denotes observed income.

Source: Albert et al. (2018).

To verify the robustness of my main results, I rerun my empirical tests using income class tercile subsamples defined as follows.

Table 10 Supplemental income class category definitions

Income class categorization	Indicative range per data
First income tercile	[₱11,285, ₱127,980]
Second income tercile	[₱127,997, ₱245,592]
Third income tercile	[₱245,600, ₱12,431,270]

Indicative range figures are annualized.

Source: FIES 2015.

My original conclusions generally hold true under these new subsampling specifications.

One notable finding is that additional permanent income results in a larger decrease in the household saving rate (Table 12) and a larger increase in consumption (Table 13) for households in the second income tercile than for those in the first. One might initially think that this may seem to go my primary findings in Table 5 and Table 8, but this seeming contradiction is explained by the fact that more of the observations in the lower-income class from the original subsample (those that drive down the marginal effects of permanent income on the household saving rate, seen in Table 5, and drive up the same effects on consumption for the lower-income class subsample, seen in Table 8) are now in the second income tercile subsample. This strengthens our

inference about the nonlinear relationship between permanent income and household saving.

Table 11 Marginal effects of observed, and permanent and transitory income on the household saving rate across income terciles

		(1) Observed	(2) Estimated
FIRST INCOME TERCILE		<i>N</i> = 13,848	<i>N</i> = 13,848
Income	Permanent	8.502 ***	8.565 ***
	Transitory		8.744 ***
Consumption	Permanent	- 11.020 ***	- 11.202 ***
	Transitory		- 10.986 ***
SECOND INCOME TERCILE		<i>N</i> = 13,848	<i>N</i> = 13,848
Income	Permanent	4.266 ***	4.522 ***
	Transitory		4.349 ***
Consumption	Permanent	- 5.479 ***	- 5.628 ***
	Transitory		- 5.425 ***
THIRD INCOME TERCILE		<i>N</i> = 13,848	<i>N</i> = 13,848
Income	Permanent	0.409 ***	1.118 ***
	Transitory		0.363 ***
Consumption	Permanent	- 0.781 ***	- 1.783 ***
	Transitory		- 0.655 ***
Bootstrapped standard errors not reported. Source: Author's estimations.		* <i>p</i> < 0.10,	** <i>p</i> < 0.05, *** <i>p</i> < 0.01

Table 12 Biased marginal effects of observed, and permanent and transitory income on the household saving rate across income terciles

		(1) Observed	(2) Estimated
FIRST INCOME TERCILE		<i>N</i> = 13,848	<i>N</i> = 13,848
Income	Permanent	-0.346 ***	-5.539 ***
	Transitory		1.058 ***
SECOND INCOME TERCILE		<i>N</i> = 13,848	<i>N</i> = 13,848
Income	Permanent	0.198 ***	- 6.572 ***
	Transitory		0.683 ***
THIRD INCOME TERCILE		<i>N</i> = 13,848	<i>N</i> = 13,848
Income	Permanent	0.108 ***	0.036
	Transitory		0.115 ***
Bootstrapped standard errors not reported. Source: Author's estimations.		* <i>p</i> < 0.10,	** <i>p</i> < 0.05, *** <i>p</i> < 0.01

Table 13 Effects of additional permanent and transitory income on different consumption categories across income terciles

	(1) Lower class	(2) Middle class	(3) Upper class
TOTAL OBSERVED	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	1.259 ***	1.971 ***	0.607 ***
Transitory income	0.700 ***	0.676 ***	0.379 ***
FOOD	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.890 ***	1.005 ***	0.114 ***
Transitory income	0.332 ***	0.243 ***	0.054 ***
HOUSING	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.086 ***	0.698 ***	0.204 ***
Transitory income	0.135 ***	0.145 ***	0.089 ***
COMMUNICATION	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.013 ***	0.052 ***	0.025 ***
Transitory income	0.012 ***	0.018 ***	0.012 ***
TRANSPORTATION	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.060 ***	0.117 ***	0.052 ***
Transitory income	0.042 ***	0.052 ***	0.031 ***
EDUCATION	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.036 ***	−0.029 ***	0.010 ***
Transitory income	0.019 ***	0.035 ***	0.024 ***
CLOTHING	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.029 ***	−0.008 ***	0.010 ***
Transitory income	0.016 ***	0.016 ***	0.010 ***
NONDURABLES	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.027 ***	0.027 ***	0.029 ***
Transitory income	0.011 ***	0.013 ***	0.023 ***
DURABLES	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.005 ***	−0.051 ***	0.017 ***
Transitory income	0.010 ***	0.029 ***	0.030 ***
HEALTH	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	−0.002	0.012	0.024 ***
Transitory income	0.024 ***	0.030 ***	0.021 ***
LEISURE	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.011 ***	0.007 ***	0.006 ***
Transitory income	0.004 ***	0.006 ***	0.006 ***
SPECIAL OCCASIONS	<i>N</i> = 13,848	<i>N</i> = 13,848	<i>N</i> = 13,848
Permanent income	0.006 ***	−0.032 ***	0.006 ***
Transitory income	0.020 ***	0.022 ***	0.012 ***

Bootstrapped standard errors not reported.

* $p < 0.10$,** $p < 0.05$,*** $p < 0.01$

Source: Author's estimations.

Annex B Supplemental tests using alternate instruments for permanent income

I test the robustness of my main results by conducting additional tests using alternate instruments for permanent income. As discussed in [Section 4](#), household consumption (Zabot & Gomes, 2021; Alan et al., 2015; Dynan et al., 2004) and real asset ownership (Bebczuk et al., 2015) are provided as additional instruments.

I consider two forms of consumption as potential instruments. I first follow the line of reasoning of the permanent income hypothesis and take total consumption as my instrument. If consumption were largely driven by permanent income, then observed consumption should suffice as an instrument so long as I understand that transitory consumption shocks and measurement errors in consumption will bias the marginal effect of permanent income downward (Zabot & Gomes, 2021; Dynan et al., 2004). Next, I consider more specific categories of consumption that are likely to be driven more by permanent income, rather than transitory. Zabot & Gomes (2021) and Alan et al. (2015) use all forms of nondurable consumption, but I take a more specific approach and single out the categories of consumption I identify to be essential and frequently incurred: food, housing, communication, and transportation. I use the sum of these four consumption categories as my alternate consumption instrument.

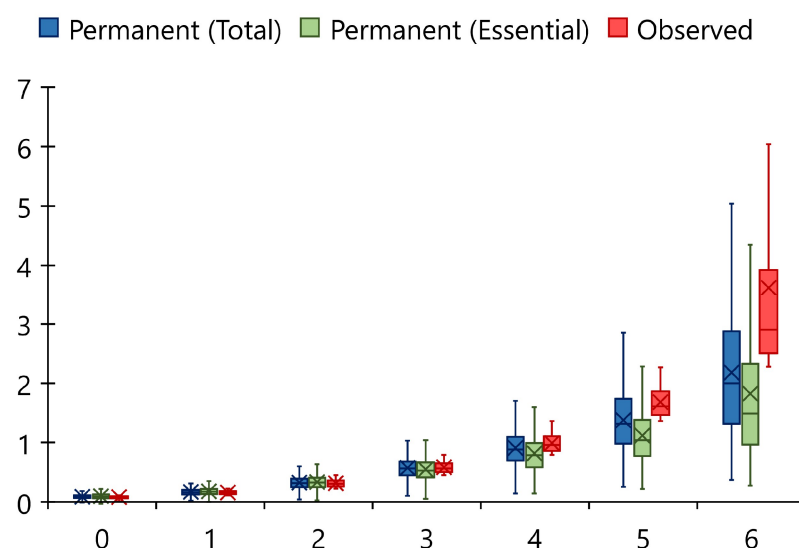
Table 14 Marginal effects of permanent and transitory income on the household saving rate under alternate specifications using consumption instruments

	(1) Total	(2) Essential
ENTIRE SAMPLE	<i>N</i> = 41,544	<i>N</i> = 41,544
Permanent	−0.034 ***	−0.063 ***
Transitory	0.522 ***	0.420 ***
LOWER CLASS	<i>N</i> = 26,341	<i>N</i> = 26,341
Permanent	−3.106 ***	−3.341 ***
Transitory	5.877 ***	4.495 ***
MIDDLE CLASS	<i>N</i> = 14,725	<i>N</i> = 14,725
Permanent	−0.274 ***	−0.310 ***
Transitory	1.204 ***	0.794 ***
UPPER CLASS	<i>N</i> = 478	<i>N</i> = 478
Permanent	−0.121 ***	−0.070 **
Transitory	0.102 ***	0.075 ***
Bootstrapped standard errors not reported.		
Source: Author's estimations.		
	* <i>p</i> < 0.10,	** <i>p</i> < 0.05, *** <i>p</i> < 0.01

I confirm that, except for those in the full-sample estimation, the coefficient estimates for permanent income decrease (reflecting the downward bias mentioned above) while the coefficient estimates for transitory income increase. I also note that, under these new specifications, the magnitude of the coefficient estimates for permanent income are closer to those for observed income (shown in [Table 5](#)) than are the coefficient estimates for transitory income. This suggests that the contribution of permanent income to the effect of observed income on the household saving rate is

greater than my original estimates suggest. From this I infer that my permanent income estimates using consumption instruments are larger than those I originally obtained using the college educational attainment of the household head. [Figure 5](#) and [Figure 6](#) confirm this inference.

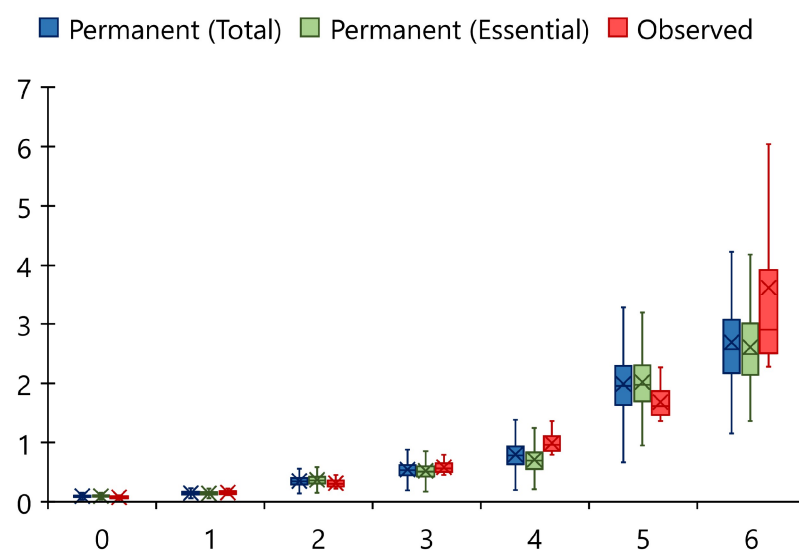
Figure 5 Full-sample permanent income estimates (using consumption instruments) vs. observed income



Units in million Philippine pesos. Outside values not displayed.

Source: Author's estimations.

Figure 6 Subsample permanent income estimates (using consumption instruments) vs. observed income



Units in million Philippine pesos. Outside values not displayed.

Source: Author's estimations.

Contrary to my primary findings, the coefficient estimates for permanent income remain negative across all subsamples, suggesting that the marginal propensities to consume out of permanent income are large enough for all households across the income distribution to cause dissaving. Though I question the validity of this proposition, I nonetheless find value in confirming that using observed income values to test the relationship between income and the household saving rate overstates the estimated effect of the former on the latter.

I also consider two additional instruments related to home ownership. First, I use home ownership alone as the instrument. (I explain why in [Section 4](#).) I also follow the reasoning behind using consumption as an instrument but narrow down my focus to regular expenses related to home ownership that may also be positively correlated with permanent income: real property tax. I intuit that using a specific portion of consumption only induces a smaller bias than using the total amount. I also attempt to strengthen the predictive power of the model for upper-class households based on the intuition that higher-income households pay more in property tax. I thus use this variable as an instrument in addition to my original instrument and home ownership.

Using both home ownership variables as instruments for permanent income seem to have almost the same effect as using the college educational attainment of the household head alone for the same purpose. [Figure 7](#) shows that the resulting permanent income estimates using home ownership instruments are also very close to my original estimates, but adding the real property tax instrument yields slightly higher permanent income estimates for upper-class households.

Table 15 Marginal effects of permanent and transitory income on the household saving rate under alternate specifications using home ownership instruments

	(1) Home ownership	(2) Property Tax
ENTIRE SAMPLE	<i>N</i> = 41,544	<i>N</i> = 41,544
Permanent	−0.053 ***	−0.021 *
Transitory	0.168 ***	0.178 ***
LOWER CLASS	<i>N</i> = 26,341	<i>N</i> = 26,341
Permanent	−2.194 ***	−2.165 ***
Transitory	0.816 ***	0.823 ***
MIDDLE CLASS	<i>N</i> = 14,725	<i>N</i> = 14,725
Permanent	−0.077 ***	−0.074 **
Transitory	0.215 ***	0.221 ***
UPPER CLASS	<i>N</i> = 478	<i>N</i> = 478
Permanent	0.085 ***	0.050 **
Transitory	0.044 ***	0.047 ***

Bootstrapped standard errors not reported.

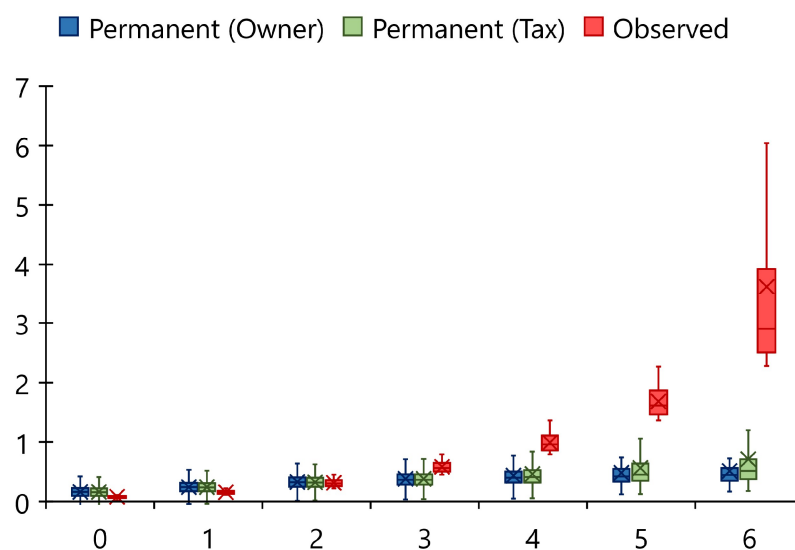
Source: Author's estimations.

* $p < 0.10$,

** $p < 0.05$,

*** $p < 0.01$

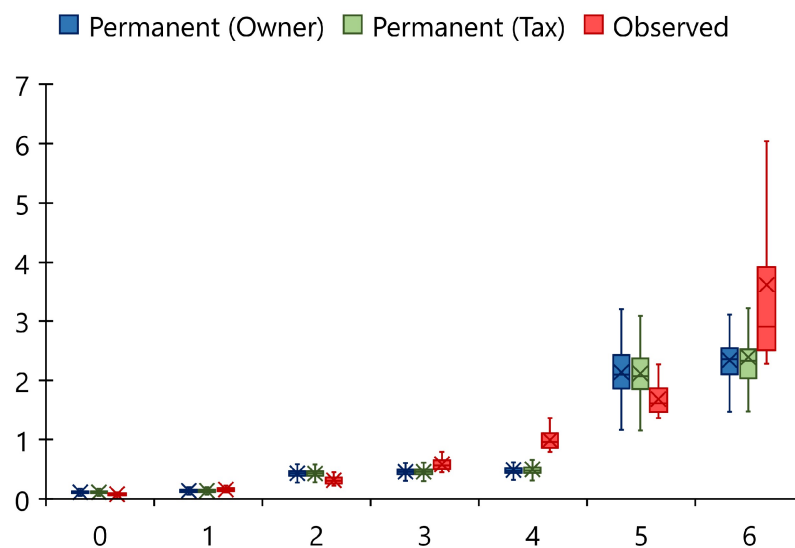
Figure 7 Full-sample permanent income estimates (using home ownership instruments) vs. observed income



Units in million Philippine pesos. Outside values not displayed.

Source: Author's estimations.

Figure 8 Subsample permanent income estimates (using home ownership instruments) vs. observed income



Units in million Philippine pesos. Outside values not displayed.

Source: Author's estimations.